Foundations & Reasoning

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Two angles, $\angle A$ and $\angle B$, are complementary. If $\angle A = (3x + 15)^\circ$ and $\angle B = (2x + 30)^\circ$, what is the measure of $\angle A$?

Complementary angles are two angles that add up to 90°. Therefore, $\angle A + \angle B = 90^\circ$. Substituting the given expressions for $\angle A$ and $\angle B$: $(3x + 15) + (2x + 30) = 90$ Combine like terms: $5x + 45 = 90$ Subtract 45 from both sides: $5x = 45$ Divide by 5: $x = 9$ Now, substitute the value of x back into the expression for $\angle A$: $\angle A = 3x + 15 = 3(9) + 15 = 27 + 15 = 42^\circ$. To verify, $\angle B = 2x + 30 = 2(9) + 30 = 18 + 30 = 48^\circ$..

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In the isosceles triangle ABC shown below, AB = AC. If the measure of angle BAC is 52°, what is the measure of angle ABC?

In triangle ABC, AB = AC, which means triangle ABC is an isosceles triangle with base BC. Therefore, the base angles ABC and ACB are equal. The sum of the interior angles of a triangle is 180°. So, $\angle BAC + \angle ABC + \angle ACB = 180^\circ$. We are given that $\angle BAC = 52^\circ$. Let $x = \angle ABC = \angle ACB$. Then, $52^\circ + x + x = 180^\circ$. $52^\circ + 2x = 180^\circ$. Subtracting 52° from both sides: $2x = 180^\circ - 52^\circ$. $2x = 128^\circ$. Dividing by 2.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In the given circle, the central angle $\angle AOB$ measures 120°. What is the measure of the inscribed angle $\angle ACB$ that subtends the same arc AB?

The relationship between a central angle and an inscribed angle that subtend the same arc is that the inscribed angle is half the measure of the central angle. Given: Central angle $\angle AOB = 120^\circ$. We want to find the inscribed angle $\angle ACB$. According to the theorem, $\angle ACB = \frac{1}{2}\angle AOB$. Substituting the given value: $\angle ACB = \frac{1}{2}(120^\circ) = 60^\circ$. Therefore, the measure of the inscribed angle $\angle ACB$ is 60°.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the line L with the equation $3x - 4y + 15 = 0$. A point P has coordinates $(5, -2)$. Calculate the shortest perpendicular distance from point P to line L.

The formula for the shortest perpendicular distance from a point (x_0, y_0) to a line $Ax + By + C = 0$ is given by $d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$. In this problem, the point P has coordinates $(x_0, y_0) = (5, -2)$ and the line L is $3x - 4y + 15 = 0$. Thus, $A = 3$, $B = -4$, and $C = 15$. Substituting these values into the distance formula: $d = \frac{|(3)(5) + (-4)(-2) + 15|}{\sqrt{3^2 + (-4)^2}}$ $d = \frac{|15 + 8 + 15|}{\sqrt{9 + 16}}$ $d = \frac{|38|}{\sqrt{25}}$ $d = \frac{38}{5}$ T.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC, points M and N are the midpoints of sides AB and AC respectively. If the area of ΔAMN is $18\ \text{cm}^2$, what is the area of trapezoid $BMNC$?

By the Midsegment Theorem, segment $MN \parallel BC$ and $MN = \frac{1}{2}BC$. Thus, $\Delta AMN \sim \Delta ABC$ with linear scale factor $k = \frac{1}{2}$. The area ratio is $k^2 = (1/2)^2 = 1/4$. Thus, $\text{Area}(\Delta ABC) = 4 \times \text{Area}(\Delta AMN) = 4 \times 18 = 72\ \text{cm}^2$. The trapezoid's area is $\text{Area}(BMNC) = 72 - 18 = 54\ \text{cm}^2$.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the figure below, lines AB and CD are parallel with $AB = 10$ and $CD = 15$. Segments AD and BC intersect at point E. If the perimeter of ΔABE is 26, what is the perimeter of ΔCDE?

Because $AB \parallel CD$, alternate interior angles make $\Delta ABE \sim \Delta DCE$. The linear scale factor between ΔCDE and ΔABE is $k = \frac{CD}{AB} = \frac{15}{10} = \frac{3}{2} = 1.5$. Since the ratio of perimeters of similar triangles equals the linear scale factor, $\text{Perimeter}(\Delta CDE) = 1.5 \times 26 = 39$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Chords AB and CD intersect inside a circle at point P. The segments on chord AB measure $PA = x$ and $PB = x + 7$. The segments on chord CD measure $PC = 3$ and $PD = 6$. What is the total length of chord AB?

By the Intersecting Chords Theorem: $PA \cdot PB = PC \cdot PD \Rightarrow x(x + 7) = 3 \times 6 = 18 \Rightarrow x^2 + 7x - 18 = 0$. Factoring: $(x + 9)(x - 2) = 0$. Since length must be positive, $x = 2$. Therefore, $PA = 2$ and $PB = 2 + 7 = 9$. The total length of chord $AB = PA + PB = 2 + 9 = 11$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC, $\angle A = 50^\circ$. Transversal DE is drawn such that $\angle ADE = \angle C = 65^\circ$. What is the measure of $\angle B$?

In ΔABC, the sum of angles is 180°: $\angle B = 180^\circ - \angle A - \angle C = 180^\circ - 50^\circ - 65^\circ = 65^\circ$. Because $\angle B = \angle C = 65^\circ$, ΔABC is an isosceles triangle with $AB = AC$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In an hourglass figure with $AB \parallel CD$, $\angle BAE = 38^\circ$ and $\angle AEB = 84^\circ$. What is the measure of $\angle EDC$?

In ΔABE, the angle sum is 180°: $\angle ABE = 180^\circ - 38^\circ - 84^\circ = 58^\circ$. Because lines AB and CD are parallel, alternate interior angles are equal: $\angle EDC = \angle ABE = 58^\circ$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A circle has a radius of 6. A sector of the circle is formed by a central angle of 60°. What is the area of the sector?

The area of a sector with radius r and central angle θ (in degrees) is given by the formula: Area $= \frac{\theta}{360^\circ}\pi r^2$ Given $r = 6$ and $\theta = 60^\circ$: Area $= \frac{60^\circ}{360^\circ}\pi(6^2) = \frac{1}{6}\pi(36) = 6\pi$ - 2π is incorrect because it is the arc length of the sector, $\frac{60^\circ}{360^\circ}(2\pi r) = \frac{1}{6}(12\pi) = 2\pi$, rather than the sector area. - 12π is incorrect because it is the full circumference of the circle.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In right triangle ABC, $\angle B = 90^\circ$, $\angle A = 60^\circ$, and the length of side AB is 5. What is the length of side BC?

In right triangle ABC, the sum of the angles is 180°, so: $\angle C = 180^\circ - 90^\circ - 60^\circ = 30^\circ$ This is a $30^\circ - 60^\circ - 90^\circ$ special right triangle. The lengths of the sides opposite the angles 30°, 60°, and 90° are in the ratio $1 : \sqrt{3} : 2$. Side AB is opposite the 30° angle, making it the shorter leg with length 5. Side BC is opposite the 60° angle, which is the longer leg. Therefore.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

On a number line, point A is at -3, and point B is at 7. Point M is the midpoint of segment $\overline{AB}$. What is the coordinate of point M?

The midpoint M of a segment with endpoints A and B on a number line is found using the formula: $M = \frac{A+B}{2}$. Given that $A = -3$ and $B = 7$, we substitute these values into the formula: $M = \frac{-3+7}{2} = \frac{4}{2} = 2$. Therefore, the coordinate of point M is 2.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the given right triangle ABC, where angle C is the right angle, the length of side AC is 5 units and the length of side BC is 5 units. What is the exact length of the hypotenuse AB?

Since triangle ABC is a right triangle with angle C = 90°, we can use the Pythagorean theorem: $a^2 + b^2 = c^2$, where a and b are the lengths of the legs and c is the length of the hypotenuse. Given AC = 5 and BC = 5, we have a = 5 and b = 5. Therefore, $5^2 + 5^2 = c^2$ $25 + 25 = c^2$ $50 = c^2$ $c = \sqrt{50}$ To simplify the radical, we find the largest perfect square factor of 50, which is 25. $c = \sqrt{25 \cdot 2}$ $c = \sqrt{25} \cdot \sqrt{2}$ $c = 5\sqrt{2}$ Alternatively, since the two legs are equal.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC, line segment DE is parallel to BC, with point D lying on $\overline{AB}$ and point E lying on $\overline{AC}$. If $AD = 3$, $DB = 6$, and $AE = 5$, what is the length of segment $\overline{EC}$?

By the Triangle Proportionality Theorem (Side-Splitter Theorem), if a line is parallel to one side of a triangle intersecting the other two sides, it divides the two sides proportionally: $\frac{AD}{DB} = \frac{AE}{EC}$ Substitute the given lengths: $\frac{3}{6} = \frac{5}{EC}$ Simplify the ratio on the left: $\frac{1}{2} = \frac{5}{EC}$ Cross-multiplying gives: $EC = 5 \times 2 = 10$ Common misconceptions: - Additive reasoning: finding $DB - AD = 6 - 3 = 3$ and computing $AE + 3 = 8$ (Choice A). - Full side length.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

In the coordinate plane, the directed line segment from $A(-5, -3)$ to $B(5, 7)$ is partitioned by point P in the ratio $3 : 2$. What are the coordinates of point P?

To find the coordinates of point P that partitions the directed segment from $A(x_1, y_1)$ to $B(x_2, y_2)$ in the ratio $m : n = 3 : 2$, we find the fraction of the total distance from A to B: $k = \frac{m}{m+n} = \frac{3}{3+2} = \frac{3}{5}$ Now, apply this fractional distance to the changes in x and y: $x_P = x_1 + k(x_2 - x_1) = -5 + \frac{3}{5}(5 - (-5)) = -5 + \frac{3}{5}(10) = -5 + 6 = 1$ $y_P = y_1 + k(y_2 - y_1) = -3 + \frac{3}{5}(7 - (-3)) = -3 + \frac{3}{5}(10) = -3 + 6 = 3$ Thus, the coordinat.

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16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A student whose eyes are 1.5 m above the ground places a mirror flat on the ground 2 m in front of her feet. Looking into the mirror, she sees the reflection of the top of a school flagpole located 18 m beyond the mirror. If the angle of incidence equals the angle of reflection, how tall is the flagpole?

The student, the ground, and the line of sight form a right triangle with legs 1.5 m (eye height) and 2 m (distance to mirror). The flagpole, ground, and reflected ray form a second right triangle with distance 18 m and height h. Because the angle of incidence equals the angle of reflection, the two right triangles are similar by AA. Setting up the proportion: $\frac{h}{1.5} = \frac{18}{2} = 9 \Rightarrow h = 9 \times 1.5 = 13.5$ m.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

In ΔABC, line segment DE is drawn parallel to base BC such that it divides the triangle into two regions of equal area: $\text{Area}(\Delta ADE) = \text{Area}(BDEC)$. What is the ratio $\frac{AD}{AB}$?

Since $DE \parallel BC$, $\Delta ADE \sim \Delta ABC$. Let $\text{Area}(\Delta ADE) = A$. Then $\text{Area}(BDEC) = A$, making the total area $\text{Area}(\Delta ABC) = 2A$. The area ratio is $\frac{\text{Area}(\Delta ADE)}{\text{Area}(\Delta ABC)} = \frac{A}{2A} = \frac{1}{2}$. Since area scales as the square of linear dimensions, $\left(\frac{AD}{AB}\right)^2 = \frac{1}{2} \Rightarrow \frac{AD}{AB} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Circles

In ΔABC, a transversal segment DE connects side AB at D and side AC at E such that $\angle ADE = \angle C$. If $AD = 3$, $AC = 9$, and the area of ΔADE is 8, what is the area of quadrilateral $BDEC$?

Because $\angle ADE = \angle C$ and both triangles share $\angle A$, $\Delta ADE \sim \Delta ACB$ by AA. The linear scale factor from the small triangle to the large triangle is $k = \frac{AC}{AD} = \frac{9}{3} = 3$. The ratio of their areas is $k^2 = 3^2 = 9$. Thus, $\text{Area}(\Delta ABC) = 9 \times \text{Area}(\Delta ADE) = 9 \times 8 = 72$. The area of quadrilateral $BDEC = \text{Area}(\Delta ABC) - \text{Area}(\Delta ADE) = 72 - 8 = 64$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Area, Surface Area & Volume

In ΔABC shown below, point D is on AB and E on AC such that $\angle ADE \cong \angle C$. If $AD = 4$, $AE = 5$, and $AC = 12$, what is the length of side AB?

Both ΔADE and ΔACB share $\angle A$, and $\angle ADE \cong \angle C$. By AA similarity, $\Delta ADE \sim \Delta ACB$ in INVERTED order! Therefore, the corresponding sides satisfy: $\frac{AD}{AC} = \frac{AE}{AB} \Rightarrow \frac{4}{12} = \frac{5}{AB} \Rightarrow \frac{1}{3} = \frac{5}{AB} \Rightarrow AB = 15$. Warning: Reflexively assuming $DE \parallel BC$ gives the incorrect distractor 9.6.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Transformations & Symmetry

ΔABC and ΔDEF are shown below with $\angle A \cong \angle D = 55^\circ$. If $AB = 6$, $AC = 10$, $DE = 9$, and $DF = 15$, which similarity criterion applies and what is the scale factor from ΔABC to ΔDEF?

Check the ratio of corresponding sides adjacent to the congruent included angle: $\frac{DE}{AB} = \frac{9}{6} = 1.5$ and $\frac{DF}{AC} = \frac{15}{10} = 1.5$. Since two pairs of corresponding sides are in the same constant ratio and their included angles are congruent ($\angle A \cong \angle D$), $\Delta ABC \sim \Delta DEF$ by SAS similarity with scale factor $k = 1.5$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 3$ in points, lines, and planes, evaluate $f(2)$.

a. 7 ✓

b. 9

c. 4

d. 11

Substitute $x = 2$: $f(2) = 2(2) + 3 = 7$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in line segments and distance, evaluate $f(2)$.

a. 10 ✓

b. 13

c. 6

d. 14

Substitute $x = 2$: $f(2) = 3(2) + 4 = 10$.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 5$ in points and midpoints, evaluate $f(2)$.

a. 13 ✓

b. 17

c. 8

d. 17

Substitute $x = 2$: $f(2) = 4(2) + 5 = 13$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for line segments and distance consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 6$ in review: slopes of lines, evaluate $f(4)$.

a. 26 ✓

b. 31

c. 20

d. 30

Substitute $x = 4$: $f(4) = 5(4) + 6 = 26$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 11

c. 9 ✓

d. 11

Substituting $x = 1$ into $y = 2x + 7$ gives $y = 2(1) + 7 = 9$.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
In ΔABC studying review: angles and angle measure, $m\angle A = 34^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 87° ✓
b. 97°
c. 79°
d. 146°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (34^\circ + 59^\circ) = 87^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length?

a. 35 cm
b. 40 cm ✓
c. 48 cm
d. 200 cm

Multiplying the original length by the scale factor gives $8 \times 5 = 40$ cm.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
A student checks that $3, 5, 7, 9$ are the first four terms of a pattern, all odd, and concludes the next term will also be odd. This is an example of:

a. Deductive reasoning
b. Circular reasoning
c. A logical fallacy
d. **Inductive reasoning** ✓

Inductive reasoning draws a general conclusion from a limited number of specific observations or patterns.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
The base angles of any isosceles triangle are congruent; this triangle is isosceles, so its base angles are congruent. This is an example of:

a. Circular reasoning
b. **Deductive reasoning** ✓
c. Inductive reasoning
d. A logical fallacy

Deductive reasoning applies a general rule or definition to a specific case to reach a certain conclusion.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A student notices the first three shapes in a sequence are all pentagons and predicts the fourth shape is also a pentagon. This is an example of:

a. A logical fallacy

b. **Inductive reasoning ✓**

c. Circular reasoning

d. Deductive reasoning

Inductive reasoning draws a general conclusion from a limited number of specific observations or patterns.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

After measuring the angles of five different triangles and finding each sums to 180°, a student concludes all triangles have angle sum 180°. This is an example of:

a. A logical fallacy

b. Circular reasoning

c. **Inductive reasoning ✓**

d. Deductive reasoning

Inductive reasoning draws a general conclusion from a limited number of specific observations or patterns.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A student measures the diagonals of four different rectangles, finds them equal each time, and concludes all rectangles have equal diagonals. This is an example of:

a. A logical fallacy

b. Deductive reasoning

c. **Inductive reasoning ✓**

d. Circular reasoning

Inductive reasoning draws a general conclusion from a limited number of specific observations or patterns.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The sum of the angles in any triangle is 180°; this triangle has angles 70° and 60°, so its third angle is 50°. This is an example of:

a. Inductive reasoning

b. Circular reasoning

c. A logical fallacy

d. **Deductive reasoning ✓**

Deductive reasoning applies a general rule or definition to a specific case to reach a certain conclusion.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Noticing that $1, 4, 9, 16$ are the first four square numbers, a student predicts the next term is 25. This is an example of:

a. Circular reasoning

b. A logical fallacy

c. **Inductive reasoning ✓**

d. Deductive reasoning

Inductive reasoning draws a general conclusion from a limited number of specific observations or patterns.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A student tests $n = 1, 2, 3$ in a formula, finds it works each time, and concludes the formula works for all n. This is an example of:

a. Circular reasoning

b. A logical fallacy

c. **Inductive reasoning ✓**

d. Deductive reasoning

Inductive reasoning draws a general conclusion from a limited number of specific observations or patterns.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Circles

What is the contrapositive of the statement: "If A triangle has a right angle, then it is a right triangle."?

a. If It is a right triangle, then a triangle has a right angle.

b. **If It is not a right triangle, then a triangle does not have a right angle. ✓**

c. If A triangle does not have a right angle, then it is not a right triangle.

d. If A triangle does not have a right angle, then it is a right triangle.

The contrapositive swaps and negates both parts — it is always logically equivalent to the original.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Area, Surface Area & Volume

Through any three noncollinear points, there is exactly one ___.

a. line

b. segment

c. ray

d. **plane** ✓

This is one of the basic postulates/definitions of Euclidean geometry: the answer is plane.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 9.01 Transformations & Symmetry

If two distinct planes intersect, their intersection is a ___.

a. ray

b. **line** ✓

c. point

d. segment

This is one of the basic postulates/definitions of Euclidean geometry: the answer is line.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Coordinate Geometry

If B is between A and C on a line, then by the Segment Addition Postulate, $AB + BC =$

a. $AC/2$

b. $2AC$

c. AC ✓

d. BC

This is one of the basic postulates/definitions of Euclidean geometry: the answer is AC.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 11.01 Probability

A line contains at least ___ points.

a. infinitely many

b. one

c. three

d. **two** ✓

This is one of the basic postulates/definitions of Euclidean geometry: the answer is two.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Lines & Angles

What is the inverse of the statement: "If A quadrilateral is a parallelogram, then its opposite sides are parallel."?

a. If A quadrilateral is a parallelogram, then its opposite sides are not parallel.

b. If Its opposite sides are parallel, then a quadrilateral is a parallelogram.

c. **If A quadrilateral is not a parallelogram, then its opposite sides are not parallel.** ✓

d. If Its opposite sides are not parallel, then a quadrilateral is not a parallelogram.

The inverse negates both the hypothesis and the conclusion, keeping their order.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Triangles & Congruence

What is the inverse of the statement: "If A shape is a rhombus, then its diagonals are perpendicular."?

a. **If A shape is not a rhombus, then its diagonals are not perpendicular.** ✓

b. If Its diagonals are not perpendicular, then a shape is not a rhombus.

c. If Its diagonals are perpendicular, then a shape is a rhombus.

d. If A shape is a rhombus, then its diagonals are not perpendicular.

The inverse negates both the hypothesis and the conclusion, keeping their order.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

What is the converse of the statement: "If A polygon is a triangle, then it has exactly three sides."?

a. **If It has exactly three sides, then a polygon is a triangle.** ✓

b. If It has exactly three sides, then a polygon is not a triangle.

c. If A polygon is not a triangle, then it does not have exactly three sides.

d. If It does not have exactly three sides, then a polygon is not a triangle.

The converse swaps the hypothesis and conclusion of the original conditional.

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